PURE 1.6.

Series

sequence = an ordered set of numbers e.g.: $a_1, a_2 \dots a_k, \dots a_n$

1, 4, 7, 10, ...

finite
- first term ✓
- last term ✓

infinite
- continues forever

1. Binomial expansion of $(a+b)^n$

"two terms"

e.g.: $1b^4$

index

coefficient

$$(a+b)^0 = 1$$
$$(a+b)^1 = 1a + 1b$$
$$(a+b)^2 = 1a^2 + 2ab + 1b^2$$
$$(a+b)^3 = 1a^3 + 3a^2b + 3ab^2 + 1b^3$$
$$(a+b)^4 = 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4$$

n = 4

- a: 4 $\xrightarrow{-1}$ 3 $\xrightarrow{-1}$ 2 $\xrightarrow{-1}$ 1 $\xrightarrow{-1}$ 0
- b: 0 $\xrightarrow{+1}$ 1 $\xrightarrow{+1}$ 2 $\xrightarrow{+1}$ 3 $\xrightarrow{+1}$ 4

4 4 4 .....

all terms have a total index of 4

= coefficients form a pattern: "Pascal's triangle"

* Method 1.

$n=0$:	1					
$n=1$:	1	1				
$n=2$:	1	2	1			
$n=3$:	1	3	3	1		
$n=4$:	1	4	6	4	1	

The next row is then:

$n=5$:	1	5	10	10	5	1

$\Rightarrow 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5$

* Method 2

r = position

${}^nC_r$ function on calculator

(more on this in Stats 5.2. Combinations)

OR $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$

e.g.: $(a+b)^5$. Find coefficient for $(2+1)^{th}$ term

→ calculator $\binom{5}{2}$

1st	2nd	3rd	4th	5th	6th
$1a^5$ +	$5a^4b$ +	$10a^3b^2$ +	$10a^2b^3$ +	$5ab^4$ +	$1b^5$
$\binom{5}{0}$	$\binom{5}{1}$				$\binom{5}{5}$

↳ coefficient = 5

1^{st} term always starts at 0

2. Binomial coefficients

= coefficients in expansion of $(1+x)^n$ — better way to deal with bigger numbers

* **Binomial theorem**: (n is a positive integer)

$$\boxed{(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n, \text{ where } (r+1)^{th} \text{ term} = \binom{n}{r}x^r}$$

or extend this rule to give:

$$\boxed{(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n, \text{ where } (r+1)^{th} \text{ term} = \binom{n}{r}a^{n-r}b^r}$$

* $\binom{n}{0} = 1$; e.g.: $\binom{5}{0} = {}^5C_0 = \frac{5!}{0!\,5!} = \frac{5!}{0} = 1$

$\binom{n}{n} = 1$; e.g.: $\binom{5}{5} = {}^5C_5 = \frac{5!}{5!\,0!} = \frac{5!}{0} = 1$

$$\binom{5}{3} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!}\,2!} = \frac{5 \times 4}{2 \times 1} = 10$$

$$\binom{8}{4} = \frac{8!}{4!\,4!} = \frac{8 \times 7 \times 6 \times 5 \times \cancel{4!}}{4!\,\cancel{4!}} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$$

↳ $$\binom{n}{r} = \frac{n \times (n-1) \times (n-2) \times \dots \times (n-r+1)}{r \times (r-1) \times (r-2) \times \dots \times 3 \times 2 \times 1}$$

3. Arithmetic progressions

= linear sequence + + +

term	1st	2nd	3rd	4th	5th
	1	4	7	10	13
	a	$a+d$	$a+2d$	$a+3d$	$a+4d$

(+3 between each consecutive term)

↳ $\boxed{n^{th} \text{ term} = a + (n-1)d}$

common difference (d)

first term (a)

l = last term

* Σ all terms in a sequence = a "series"

* fear not, for all these formulas are provided on your exams :)

$$\boxed{S_n = \frac{n}{2}(a+\ell)} \quad \text{or} \quad \boxed{S_n = \frac{n}{2}[2a + (n-1)d]}$$

e.g.: $\underbrace{1 + \underbrace{2 + \underbrace{3 + \ldots + 98}_{101} + 99}_{101} + 100}_{101} = ?$

$$= \frac{100}{2} \times (1 + 100) = 50 \times 101 = 5050$$

$$\boxed{n^{th} \text{ term} = S_n - S_{n-1}}$$

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4. Geometric progressions

term	1^{st}	2^{nd}	3^{rd}	4^{th}
	2	6	18	54
	a	ar	ar^2	ar^3

$2 \xrightarrow{\times 3} 6 \xrightarrow{\times 3} 18 \xrightarrow{\times 3} 54$

common ratio

$$\boxed{n^{th} \text{ term} = ar^{n-1}}$$

Σ geometric progression

$$\boxed{S_n = \frac{a(1-r^n)}{1-r}} \quad \text{or} \quad \boxed{S_n = \frac{a(r^n-1)}{r-1}}$$

e.g.: number series $3 + 6 + 12 + 24 + \ldots$

Find sum of first 12 terms:

$$\left.\begin{array}{l} a = 3 \\ r = 2 \\ n = 12 \end{array}\right\} \quad S_n = \frac{a(1-r^n)}{1-r} \Rightarrow S_{12} = \frac{3(1-2^{12})}{1-2} = 12\ 285$$

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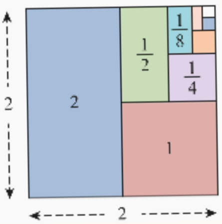
5. Infinite geometric series

= terms continue forever



e.g.: $\left.\begin{array}{l} a = 2 \\ r = \frac{1}{2} \end{array}\right\} \quad 2, \ 1, \ \frac{1}{2}, \ \frac{1}{4}, \ \frac{1}{8}$

sum $S_1 = 2, \ S_2 = 3, \ S_3 = 3\frac{1}{2}, \ S_4 = 3\frac{3}{4}, \ S_5 = 3\frac{7}{8} \ldots$

↳ gets closer to 4; "series converges to 4"

= a convergent series

$$\boxed{S_\infty = \frac{a}{1-r} \quad \text{provided that } -1 < r < 1}$$